

Mohammad Sefid | Curriculum Vitae

Department of Communications Eng. K.N.Toosi University of Technology, Tehran, Iran

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www.linkedin.com/in/mohammad-sefid/

- Master's-level research in advanced wireless communication systems for next-generation networks, focusing on innovative system design and performance analysis
- Exposure to engineering practice through industry experience in electrical systems, safety design, and installation work
- Hands-on experience with renewable energy systems, including off-grid solar installation and field deployment
- Scientific exposure to photovoltaic solar energy systems, focusing on system architectures, power conversion technologies, and research gaps

Education

- **K. N. Toosi University of Technology** **Tehran, Iran**
MSc Electrical, Electronics and Communications Engineering **2023–Present**

Notable Courses and Projects:

- Performance improvement of RIS-assisted ISAC systems (Defended)
- Machine Learning ([Code](#)).
- Numerical Optimization ([Code](#)).
- Digital Signal Processing ([Code](#)).
- Adaptive Filters ([Code](#)).

- **University of Applied Science and Technology** **Iran**
Bachelor of Science, Electrical and Electronics Engineering

Notable Projects:

- A Review of Photovoltaic (PV) Technologies: System Components, Architectures, and Future Research Directions.

- **Arak University of Technology** **Iran**
Associate Degree, Electrical and Electronics Engineering

Publications

- IRS-Assisted Cooperative NOMA with SWIPT in ISAC Systems: A Comparison of Half-Duplex and Full-Duplex Modes. [In progress]

Professional Experience

- **K. N. Toosi University of Technology** **Tehran, Iran**
Teaching Assistant **Feb 2025- Present**

- Numerical Optimization
- Stochastic Processes
- Estimation Theory
- Information Theory

- **Zamyad Co.** **Tehran, Iran**
Intern **Jul-Sep 2018**

- Developed knowledge of automotive electrical and electronic systems
- Provided electrical and electronic support for quality control laboratories and manufacturing lines
- Evaluated safety measures and contributed to the installation of grounding (earthing) systems for technical

lines

○ **Manasazan Sadid**

Intern

Tehran, Iran

Apr-June 2021

- Gained practical experience in the planning and design of renewable energy power plants
- Supported the installation of energy systems, including off-grid solar power plants
- Developed understanding of installation project management and logistics coordination

○ **Ariosunshade**

Technical content creator

Tehran, Iran

Sep 2017-Mar 2018

- Translated technical energy-related articles from English into Farsi
- Conducted literature review to identify state-of-the-art advancements in the energy sector

Technical Tools and Methods

- **Programming:** Python | MATLAB | GitHub
- **Data Analysis:** Optimization | Machine Learning
- **Writing Tools:** L^AT_EX | Microsoft Office

Conferences

- Conference on Electrical and Computer Innovations, K. N. Toosi University of Technology, Feb 2025
Presented my research on "Applications of STAR RIS in Communication Systems"

Research Interests

- Joint Optimization of Integrated Communication and Sensing (ISAC) Performance in Next-Generation Wireless Networks
- Simultaneous Wireless Information and Power Transfer (SWIPT) for Cooperative Non-Orthogonal Multiple Access (C-NOMA) Systems
- Optimization of Reconfigurable Intelligent Surfaces
- (RIS) to Enhanced Beamforming of Wireless Signals Deep Reinforcement Learning (DRL), Classic and Metaheuristic Optimization for wireless Systems

Extra-Curricular Activities and Hobbies

- Half Marathon Runner
- Recreational Cyclist
- Workout